

Trial Higher School Certificate Examination

2011



Mathematics Extension 2

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using blue or black pen
- Write your Student Number on every page
- Begin each question in a new booklet.
- All necessary working must be shown.
- Marks may be deducted for careless or poorly presented work.
- Board-approved calculators may be used.
- A list of standard integrals is included at the end of this paper.
- The mark allocated for each question is listed at the side of the question.

Total Marks –

- Attempt ALL questions.
- All questions are of equal value.

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x, \quad x > 0$

Question 1 – (15 marks) – Start a new booklet

Marks

a) If $f(x) = \cos x + i \sin x$, show that $\frac{f'(x)}{f(x)} = i$

2

b) Find

$$\int \sin^3 x \, dx$$

3

c) Using integration by parts, or otherwise, find

3

$$\int x \tan^{-1} x \, dx$$

d) (i) Find the remainder when $x^2 + 3$ is divided by $x^2 + 2x - 3$.

1

(ii) Hence, find $\int \frac{x^2+3}{x^2+2x-3} \, dx$

3

e) If $f(x) = 2 - x^2$, without the use of calculus, sketch $y = \frac{1}{f(x)}$, showing all the asymptotes and points of intersection with the axes.

3

Question 2 – (15 marks) – Start a new booklet

Marks

a) For the function $y = \frac{\log_e(x^2-2)}{1-x}$

(i) State the domain.

1

(ii) Identify all the asymptotes.

2

(iii) Find the x -intercepts.

1

(iv) Sketch the curve without using calculus.

2

b) Find all pairs of real x and y that satisfy $(x + iy)^2 = 9 - 12i$

3

c) Sketch on the Argand Diagram the locus of a point representing the complex number z if

3

$$1 \leq |z| < 2 \text{ and } \frac{\pi}{3} \leq \arg z \leq \pi$$

d) If a complex number z is a zero of $P(x) = a_1x^n + a_2x^{n-1} + \dots + a_{n-1}x + a_n$, where $a_1, a_2, \dots, a_{n-1}, a_n$ are all rational, prove that $\bar{z}$ is also a zero of this polynomial.

3

Question 3 – (15 marks) – Start a new booklet

Marks

a) Consider the ellipse $\frac{x^2}{25} + \frac{y^2}{9} = 1$

(i) Determine the eccentricity, the coordinates of the foci (S and S') and the equations of the directrices. 3

(ii) Sketch the ellipse showing all important features. 2

(iii) P is a point on the ellipse. Show that $PS + PS'$ is a constant. 2 3

(iv) Find the gradient of the tangent at $P(5 \cos \theta, 3 \sin \theta)$ and, hence, show that the equation of the tangent at P is 3

$$\frac{x \cos \theta}{5} + \frac{y \sin \theta}{3} = 1$$

b) (i) Find all the roots of $x^6 = -1$ 2

(ii) By considering the three conjugate pairs of roots from part (i), or otherwise, express $x^6 + 1$ as a product of three quadratic factors with real coefficients. 2

Question 4 – (15 marks) – Start a new booklet

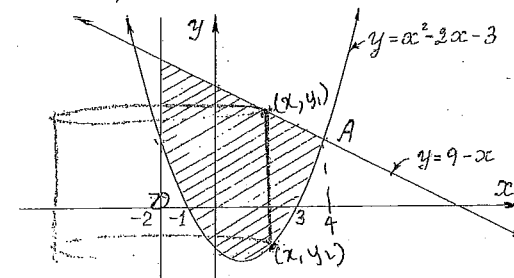
Marks

a) The area of a circle $x^2 + y^2 = r^2$ can be expressed as

$$4 \int_0^r \sqrt{r^2 - x^2} \, dx$$

Use the substitution $x = r \sin \theta$ to show that the area of the above circle is πr^2 . 3

b) The shaded region is bounded by the curve $y = x^2 - 2x - 3$, the line $y = 9 - x$ and the vertical line $x = -2$, as shown.



(i) Find the x -coordinate of point A . 1

(ii) The shaded region is rotated about the line $x = -2$. Use the method of cylindrical shells to express the volume of the resulting solid of revolution as an integral. 3

(iii) Evaluate this volume. Leave your answer in terms of π . 2

c) Use implicit differentiation to find the equation of the tangent to the curve $x^5 + 2x^2y^2 + y^3 = 2$ at the point $(1, -1)$. 3

d) $P(x) = 2x^4 + 9x^3 + 6x^2 - 20x - 24$ has a zero of multiplicity 3.

(i) Find this zero. 2

(ii) Hence factorise $P(x)$ fully. 1

Question 5 - (15 marks) - Start a new booklet

Marks

- a) (i) Prove that

$$\int_0^a f(x) dx = \int_0^a f(a-x) dx$$

2

(Hint: use substitution $u = a - x$)

(ii) Hence evaluate $\int_0^3 9x^2(3-x)^7 dx$

2

- b) Let n be an integer greater or equal to 1. Let $I_n = \int_0^1 x^n e^{-x} dx$

(i) Show that $I_{n+1} = (n+1)I_n - \frac{1}{e}$

3

(ii) Hence evaluate $\int_0^2 x^2 e^{-\frac{x}{2}} dx$ Leave your answer in exact form.

3

- c) (i) Show that if $x^4 + px + q = 0$ has a double root α , then

$$\alpha = \left(-\frac{p}{4}\right)^{\frac{1}{3}}$$

2

- (ii) Hence, show that the relationship between p and q is expressed as

$$2^8 q^3 = 3^3 p^4$$

3

Question 6 - (15 marks) - Start a new booklet

Marks

- a) (i) Sketch the curve $y = \frac{1}{x^2+1}$, clearly indicating any turning points and the behaviour of the curve as $x \rightarrow \pm\infty$

2

- (ii) Hence, on the same diagram, sketch $y = \frac{x^2}{x^2+1}$, clearly indicating any turning points, behaviour of the curve as $x \rightarrow \pm\infty$ and any points of intersection with the curve in part (i).

2

- b) At 8.10 am, at high tide, the deck of a ship was 1.6 m above the level of a wharf and at 2.30 pm, at low tide, the deck was 2.4 m below the level of the wharf. If the motion of the tide is simple harmonic:

- (i) Find when the deck was level with the wharf.

4

- (ii) Find the maximum vertical speed of the deck.

1

- c) (i) Show that $1 + \sin 2x = (\cos x + \sin x)^2$

2

- (ii) Hence, or otherwise, find all x such that

$$\cos x + \sin x = 1 + \sin 2x,$$

where $0 \leq x \leq 2\pi$

$3\frac{1}{2}$

Question 7 - (15 marks) - Start a new booklet

Marks

- a) A body of mass m kg is released from rest and falls vertically with velocity v m s⁻¹ in the medium where the resistance is $\frac{1}{10}v$. After time t seconds the body has fallen a distance of x metres.

(i) Show that the equation of the motion may be written as $\ddot{x} = g - \frac{v}{10m}$ 1

(ii) Show that the terminal velocity is given by $V_t = 10mg$ 1

(iii) Show that the time taken to reach the velocity of half the terminal velocity is $10m \ln 2$ seconds. 3

(iv) Find the distance fallen by the body when $v = \frac{1}{2}V_t$. 2 1/2

- b) The tangents of the points $P(5p, \frac{5}{p})$ and $Q(5q, \frac{5}{q})$, where $p > 0$ and $q > 0$, on the rectangular hyperbola $xy = 25$ intersect at point T .

(i) Show that the coordinates of T are $(\frac{10pq}{p+q}, \frac{10}{p+q})$. 3

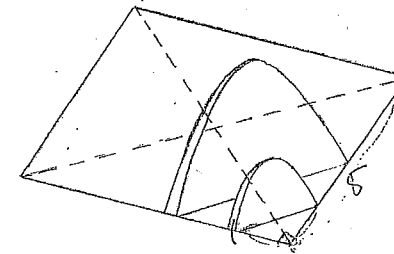
(ii) If the chord PQ produced intersects y -axis at $R(0, 5)$, show that $pq = p + q$. 2

(iii) Hence, find the locus of T and describe it geometrically. 1 1/2

Question 8 - (15 marks) - Start a new booklet

Marks

- a) (i) Show that the area enclosed by the parabola $x^2 = 4ay$ and its latus rectum, $y = a$, is $\frac{8a^2}{3}$ 4



- (ii) The base of a solid is a square of side length 5 cm and each cross-section perpendicular to the base and to one of its diagonals is the region enclosed by a parabola and its latus rectum. 3 1/2

Find the volume of this solid.

- b) AB is a common chord of two circles. A straight line through B cuts the circles at points E and F . Tangents to the circles at E and F meet at C .

- (i) Sketch a neat diagram representing the situation. 1

- (ii) Prove that $AECF$ is a cyclic quadrilateral. 3

- (iii) Show that if the circles are equal and $EB = BF$ then points A, B and C are collinear. 1 1/2

v. 1

$$f(x) = -\sin x + i \cos x$$

$$f'(x) = i(i \sin x + \cos x)$$

$$\text{So } \frac{f'(x)}{f(x)} = \frac{i(i \sin x + \cos x)}{\cos x + i \sin x} = i$$

or (II)

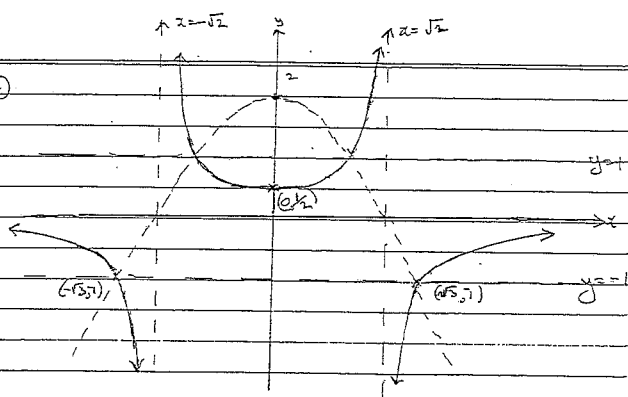
$$\frac{f'(x)}{f(x)} = \frac{-\sin x + i \cos x}{\cos x + i \sin x} \times \frac{(\cos x - i \sin x)}{(\cos x - i \sin x)} = \frac{i(\sin^2 x + \cos^2 x)}{(\cos^2 x + \sin^2 x)} = i$$

$$(b) \int \sin^3 x \, dx = \int \sin^2 x \cdot \sin x \, dx = \int (1 - \cos^2 x) \cdot \sin x \, dx$$

$$= \int \sin x \, dx - \int \cos^2 x \cdot \sin x \, dx = -\cos x + \frac{1}{3} \cos^3 x + C$$

$$(c) \int x \cdot \tan x \, dx = \left[\tan x \cdot \frac{x^2}{2} \right] - \int \frac{1}{1+x^2} \cdot \frac{x^2}{2} \, dx = \frac{x^2}{2} \tan x - \frac{1}{2} \int \frac{x^2}{1+x^2} \, dx = \frac{x^2}{2} \tan x - \frac{1}{2} \int \left(x - \frac{x}{1+x^2} \right) \, dx = \frac{x^2}{2} \tan x - \frac{1}{2} \left[x - \tan^{-1} x \right] + C = \frac{1}{2} \tan^{-1} x (x^2 + 1) - \frac{1}{2} x + C$$

(e)



(d) (i)

$$\frac{x^2 + 2x - 3}{x^2 + 2x - 3} = \frac{x^2 + 3}{x^2 + 2x - 3} - \frac{(2x - 6)}{x^2 + 2x - 3} = \frac{x^2 + 3}{x^2 + 2x - 3} - \frac{(2x - 6)}{(x^2 + 2x - 3)} = \frac{x^2 + 3}{x^2 + 2x - 3} - \frac{(2x - 6)}{(x^2 + 2x - 3)}$$

(ii) (I)

$$\int \frac{x^2 + 3}{x^2 + 2x - 3} \, dx = \int \frac{x^2 + 2x - 3 + 2(3 - x)}{x^2 + 2x - 3} \, dx = \int 1 \, dx + 2 \int \frac{3 - x}{(x+3)(x-1)} \, dx = x + 2 \int \frac{3 - x}{(x+3)(x-1)} \, dx = x + 2 \left(\frac{1}{2} \ln(x-1) - \frac{3}{2} \ln(x+3) \right) + C = x + \ln(x-1) - 3 \ln(x+3) + C$$

II

$$\int \frac{x^2 + 3}{x^2 + 2x - 3} \, dx = \int \frac{x^2 + 2x - 3 + 2(3 - x)}{x^2 + 2x - 3} \, dx = \int 1 \, dx + \int \frac{2x + 2 - 8}{x^2 + 2x - 3} \, dx = \int 1 \, dx + \int \frac{2x - 6}{(x+3)(x-1)} \, dx = x + \ln(x-1) - 2 \ln(x+3) + C = x + \ln \left(\frac{x-1}{(x+3)^2} \right) + C$$

Question 2

$$(a) y = \log_e(x^2 - 2)$$

$$(i) \text{ domain: } x^2 - 2 > 0 \Rightarrow x^2 > 2$$

$$x > \sqrt{2} \text{ or } x < -\sqrt{2}, \text{ } x \text{ is real}$$

(ii) Vertical asymptotes:

$$x^2 - 2 = 0 \Rightarrow x = \sqrt{2} \text{ and } x = -\sqrt{2}$$

Since $x = 1$ is inside the interval $-\sqrt{2} < x < \sqrt{2}$, it is not an asymptote.

Horizontal asymptote:

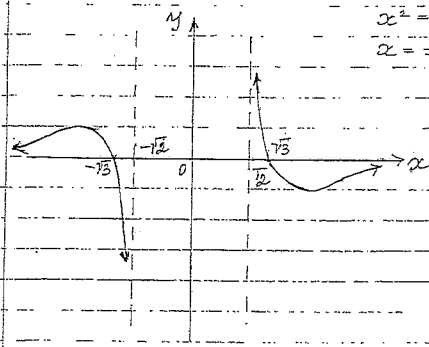
$$\lim_{x \rightarrow \infty} \frac{\log_e(x^2 - 2)}{x - x} = 0, \text{ since } x \text{ dominates } \log_e x$$

$$\lim_{x \rightarrow -\infty} \frac{\log_e(x^2 - 2)}{x - x} = 0, \text{ since } x \text{ dominates } \log_e x$$

$\therefore y = 0$ is a horizontal asymptote

$$(iii) x\text{-intercepts: } \log_e(x^2 - 2) = 0$$

$$x^2 - 2 = 1 \Rightarrow x^2 = 3 \Rightarrow x = \pm \sqrt{3}$$



$$\text{As } x \rightarrow -\sqrt{2}^-, \log_e(x^2 - 2) \rightarrow -\infty, 1 - x \rightarrow 1 + \sqrt{2} (> 0) \therefore y \rightarrow -\infty$$

$$\text{As } x \rightarrow +\sqrt{2}^-, \log_e(x^2 - 2) \rightarrow -\infty, 1 - x \rightarrow 1 - \sqrt{2} (< 0) \therefore y \rightarrow +\infty$$

(b) $(x+iy)^2 = 9-12i$
 $x^2 - y^2 = 9$
 $2xy = -12$
 $y = -\frac{6}{x}$

$x^2 - \left(-\frac{6}{x}\right)^2 = 9$

$x^4 - 36 = 9x^2$

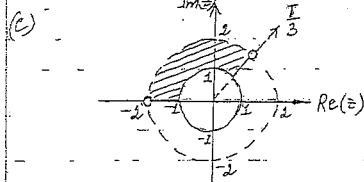
$x^4 - 9x^2 - 36 = 0$

$(x^2-12)(x^2+3)=0$

$x = \pm\sqrt{12} = \pm 2\sqrt{3}$, since x, y are real $x \neq 3$
 when $x = 2\sqrt{3}$ $y = -\frac{6}{2\sqrt{3}} = -\frac{\sqrt{3}}{1} = -\sqrt{3}$

when $x = -2\sqrt{3}$ $y = -\frac{6}{-2\sqrt{3}} = \sqrt{3}$

$\therefore x = 2\sqrt{3}, y = -\sqrt{3}$ or $x = -2\sqrt{3}, y = \sqrt{3}$



(d) $P(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0$, since z is zero.

$a_n \bar{z}^n + a_{n-1} \bar{z}^{n-1} + \dots + a_1 \bar{z} + a_0 = 0$

Taking conjugates of LHS and RHS

$a_n \bar{z}^n + a_{n-1} \bar{z}^{n-1} + \dots + a_1 \bar{z} + a_0 = 0$

$a_n \bar{z}^n + a_{n-1} \bar{z}^{n-1} + \dots + a_1 \bar{z} + a_0 = 0$

Since $\bar{\bar{z}} = z$ and

$\bar{0} = 0$

$a_n \bar{z}^n + a_{n-1} \bar{z}^{n-1} + \dots + a_1 \bar{z} + a_0 = 0$

Since $\bar{\bar{z}} = z$ and

$\bar{a} = a$ if a is real

$a_n \bar{z}^n + a_{n-1} \bar{z}^{n-1} + \dots + a_1 \bar{z} + a_0 = 0$

Since $\bar{\bar{z}} = z$

$\therefore P(\bar{z}) = 0$, as required.

QUESTION 3:

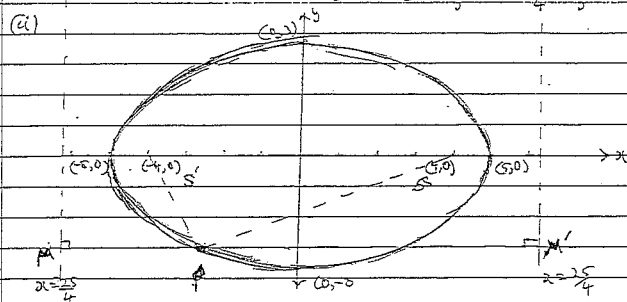
(a) $\frac{x^2}{5^2} + \frac{y^2}{3^2} = 1$ (i) $b^2 = a^2(1-e^2)$ $\therefore e^2 = 1 - \frac{b^2}{a^2}$

$e^2 = 1 - \frac{9}{25}$

$e = \frac{4}{5}$

$S(ae, 0)$; $S'(-ae, 0)$ since $a > b$
 $S(4, 0)$; $S'(-4, 0)$

Directrices $x = \pm \frac{a}{e}$ $x = \frac{25}{4}$; $x = -\frac{25}{4}$



(iii) Let $P(x, y)$ lie on the ellipse

Let M and M' be foot of perpendicular from P to directrix.

Defn $PS = e$ PM $PS' = e$ PM'

So $PS + PS' = e PM + e PM'$
 $= e (PM + PM')$
 $= e \times 2 \frac{a}{e}$
 $= 2a$

Hence, $PS + PS' = 2 \times 5$
 $= 10$ Constant

(iv) $\frac{2x}{25} + \frac{2y}{9} \frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = -\frac{9x}{25y}$

at $P(5 \cos \theta, 3 \sin \theta)$ gradient $m = -\frac{9 \times 5 \cos \theta}{25 \times 3 \sin \theta}$

$m = -\frac{3 \cos \theta}{5 \sin \theta}$

Equation of tangent is

$y - 3 \sin \theta = -\frac{3 \cos \theta}{5 \sin \theta} (x - 5 \cos \theta)$

$5y \sin \theta - 15 \sin^2 \theta = -3x \cos \theta + 15 \cos^2 \theta$

$3x \cos \theta + 5y \sin \theta = 15 (\sin^2 \theta + \cos^2 \theta)$

$\therefore \frac{x \cos \theta}{5} + \frac{y \sin \theta}{3} = 1$

(b) let $z = \cos \theta + i \sin \theta$ be a solution

to $z^6 + 1 = 0$ i.e. $z^6 = -1$

Now $|z| = 1$ $(\cos \theta + i \sin \theta)^6 = -1$

$|-1| = 1$ $\cos 6\theta + i \sin 6\theta = -1 + i0$ [De Moivre]

Equate Real & Imaginary parts.

$\cos 6\theta = -1$ $\sin 6\theta = 0$

$6\theta = 2n\pi + \pi$ $6\theta = 2n\pi + \pi$

$\theta = \frac{\pi}{6} (2n+1)$

$n=0$; $\theta = \frac{\pi}{6} \rightarrow z_1 = \cos \frac{\pi}{6} + i \sin \frac{\pi}{6}$, $z_1 = \frac{\sqrt{3}}{2} + i \frac{1}{2}$

$n=1$; $\theta = \frac{\pi}{2} \rightarrow z_2 = \cos \frac{\pi}{2} + i \sin \frac{\pi}{2}$, $z_2 = i$

$n=2$; $\theta = \frac{5\pi}{6} \rightarrow z_3 = \cos \frac{5\pi}{6} + i \sin \frac{5\pi}{6}$, $z_3 = -\frac{\sqrt{3}}{2} + i \frac{1}{2}$

$n=3$; $\theta = \pi \rightarrow z_4 = \cos \pi + i \sin \pi$, $z_4 = -1$

$n=4$; $\theta = \frac{7\pi}{6} \rightarrow z_5 = \cos \frac{7\pi}{6} + i \sin \frac{7\pi}{6}$, $z_5 = -\frac{\sqrt{3}}{2} - i \frac{1}{2}$

$n=5$; $\theta = \frac{3\pi}{2} \rightarrow z_6 = \cos \frac{3\pi}{2} + i \sin \frac{3\pi}{2}$, $z_6 = -i$

Note : $z_6 = \bar{z}_1$; $z_5 = \bar{z}_2$; $z_4 = \bar{z}_3$ Conjugate pairs

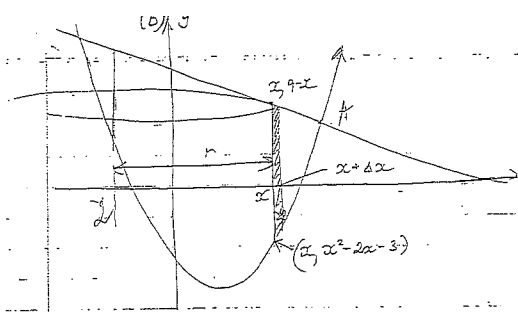
(a) $x^6 + 1 = (x-z_1)(x-z_2)(x-z_3)(x-z_4)(x-z_5)(x-z_6)$
 $= (x-z_1)(x-\bar{z}_1)(x-z_2)(x-\bar{z}_2)(x-z_3)(x-\bar{z}_3)$
 $= [x^2 - 2 \operatorname{Re}(z_1)x + |z_1|^2] [x^2 - 2 \operatorname{Re}(z_2)x + |z_2|^2] [x^2 - 2 \operatorname{Re}(z_3)x + |z_3|^2]$
 $= (x^2 - \sqrt{3}x + 1)(x^2 + \sqrt{3}x + 1)(x^2 + 1)$

Question 4

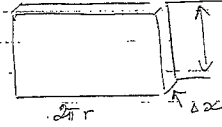
(a) Area = $4 \int_0^{\frac{\pi}{2}} \sqrt{r^2 - x^2} dx$

Let $x = r \sin \theta$
 $dx = r \cos \theta d\theta$
 when $x=0$ $\theta=0$
 $x=r$ $\theta = \frac{\pi}{2}$

$$\begin{aligned} \therefore \text{Area} &= 4 \int_0^{\frac{\pi}{2}} \sqrt{r^2 - r^2 \sin^2 \theta} r \cos \theta d\theta \\ &= 4 \int_0^{\frac{\pi}{2}} r \sqrt{1 - \sin^2 \theta} r \cos \theta d\theta \\ &= 4r^2 \int_0^{\frac{\pi}{2}} \cos^2 \theta d\theta \\ &= 4r^2 \int_0^{\frac{\pi}{2}} \frac{1}{2} (1 + \cos 2\theta) d\theta \\ &= 2r^2 \left[\theta + \frac{\sin 2\theta}{2} \right]_0^{\frac{\pi}{2}} \\ &= 2r^2 \left[\left(\frac{\pi}{2} + \frac{\sin \pi}{2} \right) - \left(0 + \frac{\sin 0}{2} \right) \right] \\ &= 2r^2 \left(\frac{\pi}{2} - 0 \right) \\ &= \pi r^2, \text{ as required} \end{aligned}$$



$r = x + 2$
 $h = (x+2) - (x^2 - 2x - 3)$
 $= 12 + x - x^2$



$\Delta V = 2\pi r h \Delta x$
 $= 2\pi (x+2) (12+x-x^2) \Delta x$
 $= 2\pi (12x + x^2 - x^3 + 24 + 2x - 2x^2) \Delta x$
 $= 2\pi (14x - x^2 - x^3 + 24) \Delta x$

$V \approx \sum_{x=0}^4 2\pi (14x - x^2 - x^3 + 24) \Delta x$

$V = 2\pi \int_0^4 (14x - x^2 - x^3 + 24) dx$
 $= 2\pi \left[7x^2 - \frac{1}{3}x^3 - \frac{1}{4}x^4 + 24x \right]_0^4$
 $= 2\pi \left(112 - \frac{64}{3} - 64 + 96 \right) - (28 + \frac{8}{3} - 4)$
 $= 2\pi \left(\frac{368}{3} + \frac{64}{3} \right)$
 $= 288\pi u^3$

(c)

$x^5 + 2x^2y^2 + y^3 = 2$

$5x^4 + (y^2 \cdot 4x + 2x^2 \cdot 2y \cdot \frac{dy}{dx}) + 3y^2 \frac{dy}{dx} = 0$

$5x^4 + 4xy^2 + \frac{dy}{dx} (4x^2y + 3y^3) = 0$

$\frac{dy}{dx} = - \frac{(5x^4 + 4xy^2)}{4x^2y + 3y^3}$

At $P(1, -1)$ $\frac{dy}{dx} = - \frac{5+4}{-4+3} = \frac{9}{-1} = -9$

Tangent has gradient 9.

$y+1 = 9(x-1)$

$y+1 = 9x-9$

$y = 9x-10$ or $9x-y-10=0$

(d) $P(x) = 2x^4 + 9x^3 + 6x^2 - 20x - 24$

$P'(x) = 8x^3 + 27x^2 + 12x - 20$

$P''(x) = 24x^2 + 54x + 12$

$P''(x) = 0$ if triple root

$24x^2 + 54x + 12 = 0$

$4x^2 + 9x + 2 = 0$

$(4x+1)(x+2) = 0$

$x = -\frac{1}{4}$ or $x = -2$

$P(-2) = -64 + 108 - 24 - 20 = 0$

$P'(-2) = -64 + 108 - 20 = 24 \neq 0$

$\therefore x = -2$ is the triple root

$P(x) = (x+2)^3 (2x-3)$

$8C = 24$

$C = -3 \Rightarrow P(x) = (x+2)^3 (2x-3)$

Question 5:

(a) let $u = a-x$ then $dx = -du$

when $x=a$, $u=0$

$x=0$, $u=a$

$\therefore \int_0^a f(a-x) dx = \int_a^0 f(u) \cdot (-du)$
 $= - \int_a^0 f(u) \cdot du$
 $= \int_0^a f(u) du$

This is $\int_0^a f(x) dx$ [since "u" any dummy variable]

(ii) $\int_0^3 9x^2 (3-x)^2 dx = \int_0^3 9(3-x)^2 [3-(3-x)]^2 dx$ from (i)
 $= \int_0^3 9(9-6x+x^2) \cdot x^2 dx$
 $= 9 \int_0^3 9x^2 - 6x^3 + x^4 dx$
 $= 9 \left[\frac{9}{3}x^3 - \frac{6}{4}x^4 + \frac{1}{5}x^5 \right]_0^3$
 $= 9 \left[3^3 \left(\frac{9}{3} - 2 + \frac{1}{5} \cdot 3^2 \right) - 0 \right]$
 $= 3^{10} \left(\frac{45}{40} + \frac{36}{40} \right)$
 $= \frac{1}{40} 3^{10} \left[\frac{59049}{40} \right]$

(b) (i) $I_n = \int_0^1 x^n \cdot e^{-x} dx$

(I) $I_n = \left[e^{-x} \frac{x^{n+1}}{n+1} \right]_0^1 - \int_0^1 \frac{x^{n+1}}{n+1} \cdot (-e^{-x}) dx$

$I_n = \frac{1}{n+1} e^{-1} + \frac{1}{n+1} \int_0^1 x^{n+1} \cdot e^{-x} dx$

$$\therefore (n+1) I_n = \frac{1}{e} + I_{n+1}$$

$$\text{So } I_{n+1} = (n+1) I_n - \frac{1}{e}$$

$$\text{or (II) now } I_{n+1} = \int_0^1 x^{n+1} \cdot e^{-x} dx$$

$$= [x^{n+1} \cdot (-e^{-x})]_0^1 - \int_0^1 (n+1)x^n \cdot (-e^{-x}) dx$$

$$I_{n+1} = -e^{-1} + (n+1) \int_0^1 x^n \cdot e^{-x} dx$$

$$I_{n+1} = (n+1) I_n - \frac{1}{e}$$

$$\text{(ii) Let } u = \frac{x}{2} \quad \text{So } I_2 = \int_0^1 x^2 \cdot e^{-\frac{x}{2}} dx$$

$$2u = x$$

$$\text{Then } 2du = dx$$

$$\text{When } x=0, u=0$$

$$x=2, u=1$$

$$\text{Sub. } I_2 = \int_0^1 (2u)^2 \cdot e^{-u} \cdot 2 du$$

$$= 8 \int_0^1 u^2 \cdot e^{-u} du$$

$$\text{Now } U_2 = \int_0^1 u^2 \cdot e^{-u} du$$

$$= 2 U_1 - \frac{1}{e}$$

$$= 2 [U_0 - \frac{1}{e}] - \frac{1}{e}$$

$$\text{So } U_0 = \int_0^1 e^{-u} du = 2 \left[(1 - \frac{1}{e}) - \frac{1}{e} \right] - \frac{1}{e}$$

$$= [-e^{-u}]_0^1 = 2 - \frac{2}{e}$$

$$= -\frac{1}{e} + 1$$

$$\text{Grns } I_2 = 8 U_2$$

$$= 8 \left[2 - \frac{2}{e} \right]$$

$$= \frac{16e - 40}{e}$$

Cube both sides of equation

$$\left(-\frac{p}{4}\right) \left(\frac{3p}{4}\right)^3 = -q^3 \quad (x-1)$$

$$\frac{3^3 p^4}{4^4} = q^3$$

$$\frac{3^3 p^4}{4^4} = 4^3 q^3 \quad [27p^4 = 256q^3]$$

$$\frac{3^3 p^4}{4^4} = 2^6 q^3$$

$$\text{or II Substitute: } \left[\left(-\frac{p}{4}\right)^{\frac{1}{3}} + p \left(\frac{p}{4}\right)^{\frac{1}{3}} \right] + q = 0$$

$$\frac{p^{\frac{1}{3}}}{4^{\frac{1}{3}}} - \frac{p^{\frac{1}{3}}}{4^{\frac{1}{3}}} + q = 0$$

$$p^{\frac{1}{3}} \left(\frac{1}{4^{\frac{1}{3}}} - \frac{1}{4^{\frac{1}{3}}} \right) = -q$$

$$p^{\frac{1}{3}} \left(-\frac{3}{4^{\frac{1}{3}}} \right) = -q$$

Cube both sides

and x-1

$$p^4 \cdot \frac{3^3}{4^4} = q^3$$

$$\frac{3^3 p^4}{4^4} = 4^3 q^3$$

$$\text{or (II) } I_2 = \int_0^2 x^2 \cdot e^{-\frac{x}{2}} dx \quad u = \frac{x}{2} ; dv = e^{-\frac{x}{2}}$$

$$du = \frac{1}{2} dx ; v = -2e^{-\frac{x}{2}}$$

$$\text{So } I_2 = [x^2 \cdot (-2e^{-\frac{x}{2}})]_0^2 - \int_0^2 2x \cdot (-2e^{-\frac{x}{2}}) dx$$

$$= \left(-\frac{8}{e}\right) + 4 \int_0^2 x \cdot e^{-\frac{x}{2}} dx$$

$$= \left(-\frac{8}{e}\right) + 4 \left[[x \cdot (-2e^{-\frac{x}{2}})]_0^2 - \int_0^2 (-2e^{-\frac{x}{2}}) dx \right]$$

$$= -\frac{8}{e} + 4 \left[\left(-\frac{4}{e}\right) + 2 \left[-2e^{-\frac{x}{2}}\right]_0^2 \right]$$

$$= -\frac{8}{e} + 4 \left[-\frac{4}{e} + 2 \left(-\frac{2}{e} + 2\right) \right]$$

$$= -\frac{40}{e} + 16 \rightarrow \frac{16e - 40}{e}$$

(C) (i) If α is a double root of $P(x)$
then $P(\alpha) = 0$ and $P'(\alpha) = 0$

$$\text{Now } P'(x) = 4x^3 + p$$

$$\text{So } P'(\alpha) = 4\alpha^3 + p = 0$$

$$\alpha^3 = -\frac{p}{4}$$

$$\alpha = \left(-\frac{p}{4}\right)^{\frac{1}{3}}$$

(ii) (I) Now on substitution

$$x^4 + px + q = 0$$

$$\text{Then } \alpha (\alpha^3 + p) = -q$$

$$\left(\frac{p}{4}\right)^{\frac{1}{3}} \left[-\frac{p}{4} + p\right] = -q$$

$$\left(-\frac{p}{4}\right)^{\frac{1}{3}} \left(\frac{3p}{4}\right) = -q$$

Question 6

$$(a) (i) y = \frac{-1}{x^2+1}$$

$$\frac{dy}{dx} = -\frac{2x}{(x^2+1)^2} \cdot dx = -\frac{2x}{(x^2+1)^2}$$

Stationary points $\frac{dy}{dx} = 0 \rightarrow x=0, y=1$

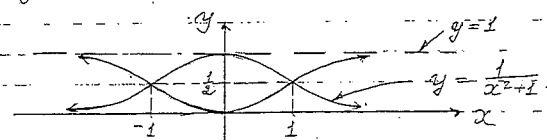
(0,1) is a stationary point

For $x > 0$ $\frac{dy}{dx} < 0$

For $x < 0$ $\frac{dy}{dx} > 0$

$\therefore (0,1)$ is a maximum turning point

as $x \rightarrow \infty y \rightarrow 0^+$ and as $x \rightarrow -\infty y \rightarrow 0^+$
 $y=0$ is a horizontal asymptote.



$$(ii) y = \frac{x^2}{x^2+1} = \frac{x^2+1-1}{x^2+1} = 1 - \frac{1}{x^2+1}$$

which is the graph from part (i) reflected in x -axis and translated 1 unit up.

$$\text{Points of intersection: } \frac{1}{x^2+1} = 1 - \frac{1}{x^2+1}$$

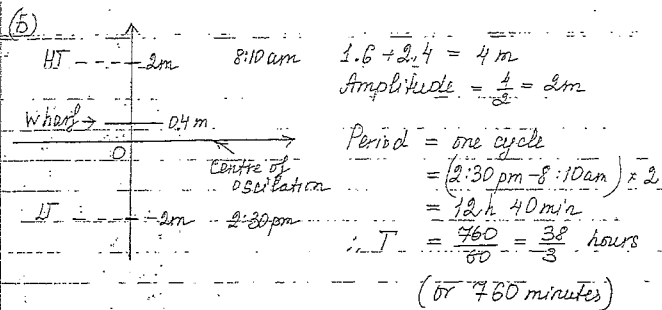
$$\frac{2}{x^2+1} = 1$$

$$x^2+1 = 2$$

$$x^2 = 1$$

$$x = -1 \quad y = \frac{1}{2}$$

$$\text{or } x = 1 \quad y = \frac{1}{2}$$



The motion is a SHM

$$x = -n^2 x, \text{ where } n = \frac{2\pi}{T} = \frac{6\pi}{38} = \frac{3\pi}{19}$$

$$(\text{or } n = \frac{2\pi}{760} = \frac{\pi}{380})$$

$$x = 2 \cos(n\pi t + \phi)$$

Initial condition: when $t=0$ $x=2$ (highest point)

$$\therefore 2 = 2 \cos \phi \Rightarrow \cos \phi = 1 \Rightarrow \phi = 0$$

$$\therefore x = 2 \cos\left(\frac{3\pi}{19}t\right) \text{ or } x = 2 \cos\left(\frac{\pi}{380}t\right)$$

$$v = \dot{x} = -\frac{3\pi}{19} \times 2 \sin\left(\frac{3\pi}{19}t\right) \quad v = \dot{x} = -\frac{\pi}{380} \times 2 \sin\left(\frac{\pi}{380}t\right)$$

$$v = -\frac{6\pi}{19} \sin\left(\frac{3\pi}{19}t\right) \quad v = -\frac{\pi}{190} \sin\left(\frac{\pi}{380}t\right)$$

Deck is level with the wharf when $x = 0.4 = \frac{2}{5}$

$$2 \cos\left(\frac{3\pi}{19}t\right) = \frac{2}{5} \text{ or } 2 \cos\left(\frac{\pi}{380}t\right) = \frac{2}{5}$$

$$\cos\left(\frac{3\pi}{19}t\right) = \frac{1}{5} \quad \cos\left(\frac{\pi}{380}t\right) = \frac{1}{5}$$

$$\frac{3\pi}{19}t = \cos^{-1}\left(\frac{1}{5}\right) \quad \frac{\pi}{380}t = \cos^{-1}\left(\frac{1}{5}\right)$$

$$t = \frac{19 \cos^{-1}\left(\frac{1}{5}\right)}{3\pi} \quad t = \frac{380 \cos^{-1}\left(\frac{1}{5}\right)}{\pi}$$

$$\approx 2.461 \text{ h} \approx 166 \text{ minutes}$$

$$\approx 166 \text{ minutes}$$

$$166 \text{ minutes} = 2 \text{ h } 46 \text{ minutes}$$

$$8:10 + 2 \text{ h } 46 \text{ min} = 10:56 \text{ am}$$

The deck is level with the wharf at 10:56 am

$$(ii) \text{ max speed when } \sin\left(\frac{3\pi}{19}t\right) = \pm 1 \text{ or } \sin\left(\frac{\pi}{380}t\right) = \pm 1$$

$$|v_{\max}| = \frac{6\pi}{19} \text{ m h}^{-1} \text{ or } |v_{\max}| = \frac{\pi}{190} \text{ m min}^{-1}$$

$$(c) \text{ RTP } 1 + \sin 2x = (\cos x + \sin x)^2$$

$$(i) \text{ RHS} = \cos^2 x + 2 \cos x \sin x + \sin^2 x$$

$$= 1 + 2 \cos x \sin x$$

$$= 1 + \sin 2x$$

$$= \text{LHS}$$

$$(ii) \cos x + \sin x = 1 + \sin 2x$$

$$\cos x + \sin x = (\cos x + \sin x)^2 \text{ (using part (i))}$$

$$\cos x + \sin x - (\cos x + \sin x)^2 = 0$$

$$(\cos x + \sin x)(1 - (\cos x + \sin x)) = 0$$

$$\cos x + \sin x = 0 \text{ or } \cos x + \sin x = 1 \text{ (since if } AB=0, \text{ then } A=0 \text{ or } B=0)$$

$$1 + \tan x = 0$$

$$\tan x = -1$$

$$x = \frac{3\pi}{4}, \frac{7\pi}{4}$$

$$\text{LHS} = \cos x + \sin x$$

$$= R \sin(x + \lambda)$$

$$= R \cos \lambda \sin x + R \sin \lambda \cos x$$

$$R \cos \lambda = 1$$

$$R \sin \lambda = 1$$

$$R^2(\sin^2 \lambda + \cos^2 \lambda) = 2$$

$$R = \sqrt{2}$$

$$\sin \lambda = \frac{1}{\sqrt{2}}, \cos \lambda = \frac{1}{\sqrt{2}}$$

$$\lambda = \frac{\pi}{4}$$

$$\therefore \sqrt{2} \sin\left(x + \frac{\pi}{4}\right) = 1$$

$$\sin\left(x + \frac{\pi}{4}\right) = \frac{1}{\sqrt{2}}$$

$$\left(x + \frac{\pi}{4}\right) = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}; x = 0, \frac{\pi}{2}, \frac{3\pi}{2}, \frac{7\pi}{2}$$

Solution:

$$x = 0, \frac{\pi}{2}, \frac{3\pi}{2}, \frac{7\pi}{2}, 2\pi$$

(c)(ii) Alternative solution

$$\cos x + \sin x = 1 + \sin 2x$$

$$(1 + \sin 2x)^2 = 1 + \sin 2x \text{ (using part (i))}$$

Squaring both sides:

$$1 + \sin 2x = 1 + 2 \sin 2x + (\sin 2x)^2$$

$$(\sin 2x)^2 + \sin 2x = 0$$

$$\sin 2x (\sin 2x + 1) = 0$$

$$\sin 2x = 0 \text{ or } \sin 2x = -1$$

$$2x = 0, \pi, 2\pi, 3\pi, 4\pi \quad 2x = \frac{3\pi}{2}, \frac{7\pi}{2}$$

$$x = 0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}, 2\pi \quad x = \frac{3\pi}{4}, \frac{7\pi}{4}$$

Now, since we squared both sides we have increased the number of solutions

$\therefore$ we need to check the obtained solutions to satisfy the original equation

$$\text{when } x=0 \quad \text{LHS}=1 \quad \text{RHS}=1 \quad \checkmark$$

$$x=\frac{\pi}{2} \quad \text{LHS}=1 \quad \text{RHS}=1 \quad \checkmark$$

$$x=\pi \quad \text{LHS}=-1 \quad \text{RHS}=1 \quad \times$$

$$x=\frac{3\pi}{2} \quad \text{LHS}=-1 \quad \text{RHS}=1 \quad \times$$

$$x=2\pi \quad \text{LHS}=1 \quad \text{RHS}=1 \quad \checkmark$$

$$x=\frac{3\pi}{4} \quad \text{LHS}=0 \quad \text{RHS}=0 \quad \checkmark$$

$$x=\frac{7\pi}{4} \quad \text{LHS}=0 \quad \text{RHS}=0 \quad \checkmark$$

Set of solutions: $x = 0, \frac{\pi}{2}, \frac{3\pi}{2}, \frac{7\pi}{2}, 2\pi$

Question 7:

(a)

At $t=0$, $x=0$ and $v=0$

(i)

$$F = mg - R$$

$$m\ddot{x} = mg - \frac{1}{10}v$$

$$\ddot{x} = g - \frac{v}{10m}$$

(ii) Terminal velocity when $\ddot{x}=0$

Let V_t be terminal velocity

$$\text{then } 0 = g - \frac{V_t}{10m}$$

$$V_t = 10mg$$

(iii) Using $\ddot{x} = \frac{dv}{dt}$

$$\text{then } \frac{dv}{dt} = g - \frac{v}{10m}$$

$$= 10mg - \frac{v}{10m}$$

$$\therefore \frac{dt}{dv} = \frac{10m}{V_t - v} \quad [V_t = 10mg]$$

$$dt = \frac{10m}{V_t - v} dv$$

$$t = \int_0^{V_t} \frac{10m}{V_t - v} dv$$

$$= -10m \left[\ln(V_t - v) \right]_0^{V_t}$$

$$= -10m \left[\ln(V_t) - \ln(V_t) \right]$$

$$= -10m \left[\ln\left(\frac{1}{2}\right) \right]$$

$$= 10m \ln 2$$

$$\text{or } \frac{dv}{dt} = \frac{10mg - v}{10m}$$

$$dt = \frac{10m}{10mg - v} dv \quad [10mg = V_0]$$

$$= \frac{10m}{V_0 - v}$$

$$\text{So } t = -10m \ln(V_0 - v) + C$$

$$\text{at } t=0, v=0 \quad \therefore 0 = -10m \ln(V_0) + C$$

$$C = 10m \ln(V_0)$$

$$t = -10m \ln \left[\frac{V_0 - v}{V_0} \right]$$

$$\text{If } v = \frac{V_0}{2}, \text{ then } t = -10m \ln \left(\frac{V_0 - \frac{V_0}{2}}{V_0} \right)$$

$$= -10m \ln \left(\frac{1}{2} \right)$$

$$= 10m \ln 2$$

$$(IV) (F) \text{ let } v \cdot \frac{dx}{dz} = \frac{10gm - v}{10m}$$

$$\frac{dv}{dz} = \frac{10gm - v}{10mv}$$

$$\frac{dx}{dz} = \frac{10mv}{V_0 - v} \quad (10gm = V_0)$$

$$= -10m \frac{(V_0 - v) + 10m V_0}{V_0 - v}$$

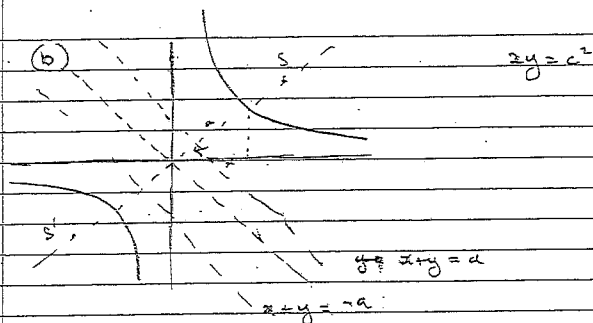
$$dx = -10m + \frac{10m V_0}{V_0 - v} dv$$

So

$$x = -10m v - 10m V_0 \ln(V_0 - v) + C$$

$$\text{at } x=0, v=0$$

$$\text{gives } C = 10m V_0 \ln(V_0)$$



$$(i) \quad xy = 25 \quad \text{at } P(S_p, \frac{5}{p})$$

$$y + x \cdot \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = -\frac{y}{x}$$

and equation of tangent

$$y - \frac{5}{p} = -\frac{1}{p} (x - 5p)$$

$$p^2 y - 5p = -x + 5p$$

$$x + p^2 y = 10p \quad \text{--- (A)}$$

Similarly for Q(S_q, \frac{5}{q})

$$\text{tangent has eq. } x + q^2 y = 10q \quad \text{--- (B)}$$

$$\text{Solve (A)-(B)} \quad (p^2 - q^2)y = 10(p - q)$$

$$y = \frac{10}{p + q}$$

$$\text{So } q^2(A) - p^2(B) \Rightarrow q^2 x + p^2 q^2 y = 10p q^2$$

$$p^2 x + p^2 q^2 y = 10q^2 p^2$$

$$(q^2 - p^2)x = 10pq(q - p)$$

$$x = \frac{10pq}{p + q}$$

$$\text{Co-ord of T } \left(\frac{10pq}{p + q}, \frac{10}{p + q} \right)$$

$$x = -10m v - 10m V_0 \ln \left[\frac{V_0 - v}{V_0} \right]$$

$$\text{take } v = \frac{V_0}{2}; x = -10m \frac{V_0}{2} - 10m V_0 \ln \left(\frac{1}{2} \right)$$

$$= 10m V_0 \left(\ln 2 - \frac{1}{2} \right)$$

$$\text{Distance fallen} = 100m^2 g \left(\ln 2 - \frac{1}{2} \right)$$

$$\text{or (II)} \quad t = -10m \ln \left[\frac{V_0 - v}{V_0} \right]$$

$$-\frac{t}{10m} = \ln \left[\frac{V_0 - v}{V_0} \right]$$

$$e^{-\frac{t}{10m}} = \frac{V_0 - v}{V_0}$$

$$V_0 - v = V_0 e^{-\frac{t}{10m}}$$

$$v = V_0 - V_0 e^{-\frac{t}{10m}}$$

$$\frac{dx}{dt} = V_0 (1 - e^{-\frac{t}{10m}})$$

$$x = V_0 t + 10m V_0 e^{-\frac{t}{10m}} + C$$

$$\text{When } t=0, x=0$$

$$\therefore C = -10m V_0$$

$$\text{So } x = V_0 t + 10m V_0 e^{-\frac{t}{10m}} - 10m V_0$$

$$\text{When } t = 10m \ln 2$$

$$x = 10mg \cdot 10m \ln 2 + 10m V_0 e^{-\ln 2} - 10m V_0$$

$$= 100m^2 g \ln 2 + 10m V_0 \frac{1}{2} - 10m V_0$$

$$*V_0 = 10mg$$

$$= 100m^2 g \ln 2 - \frac{1}{2} 100m^2 g$$

$$(i) \text{ Chord } PQ \rightarrow \text{ gradient } m = \frac{\frac{5}{p} - \frac{5}{q}}{5p - 5q}$$

$$= \frac{5(q - p)}{5(p - q)}$$

$$= -\frac{1}{pq}$$

$$\text{Equation of chord } y - \frac{5}{p} = -\frac{1}{pq} (x - 5p)$$

$$pqy - 5q = -x + 5p$$

$$x + pqy = 5(p + q)$$

$$\text{Sub } R(0,5)$$

$$0 + 5pq = 5(p + q)$$

$$pq = p + q$$

$$(ii) \text{ Now } x = \frac{10pq}{p + q}, y = \frac{10}{p + q}$$

$$\text{then } x = 10 \frac{(p+q)}{p+q} \text{ and } y = \frac{10}{p+q} \quad p > 0, q > 0$$

$$x = 10$$

$$\text{Locus is vertical line } x = 10$$

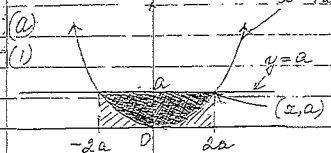
$$\text{and when } x = 10 \quad y = \frac{5}{p}$$

$$\text{Line intersects hyperbola at } (10, \frac{5}{p})$$

$$\therefore \text{ locus is vertical line } x = 10$$

$$\text{with range } 0 < y < \frac{5}{2}$$

Question 8



Method 1

$$\begin{aligned} \text{Area} &= 4a \times a - \int_{-2a}^{2a} \frac{x^2}{4a} dx \\ &= 4a^2 - \frac{1}{4a} \left[\frac{x^3}{3} \right]_{-2a}^{2a} \\ &= 4a^2 - \frac{1}{4a} \left(\frac{8a^3}{3} - 0 \right) \\ &= 4a^2 - \frac{2a^2}{3} \\ &= \frac{12a^2 - 2a^2}{3} = \frac{10a^2}{3} \end{aligned}$$

Method 2

$x^2 = 4ay \Rightarrow x = 2\sqrt{ay}$

$$\begin{aligned} \text{Area} &= 2 \int_0^a 2\sqrt{ay} dy \\ &= 4 \int_0^a \sqrt{ay} dy \\ &= 4 \left[\frac{2\sqrt{ay} \cdot y}{3} \right]_0^a \\ &= \frac{8}{3} \left[\frac{(a^2)^{3/2}}{2} - 0 \right] \\ &= \frac{8}{3} \times \frac{a^3}{2} = \frac{4a^3}{3} \end{aligned}$$

as required.

Method 3

Using the Simpson's rule:

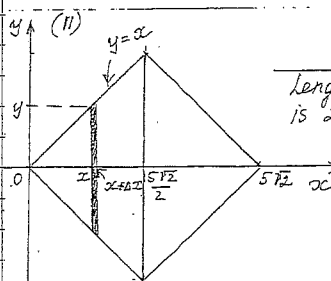
$$\int_a^b f(x) dx \approx \frac{1}{6}(b-a) \left(f(a) + 4f\left(\frac{a+b}{2}\right) + f(b) \right)$$

(note that this rule gives exact value for quadratics)

Area =

$$\begin{aligned} &\frac{1}{6}(2a - (-2a)) \left(f(-2a) + 4f(0) + f(2a) \right) \\ &= \frac{4a}{6} \times 4a = \frac{16a^2}{3} \end{aligned}$$

as required.



Length of latus rectum of a cross-section is $2y$ or $2x$, since $y = x$.

From part (i): $4a = 2x \Rightarrow a = \frac{x}{2}$

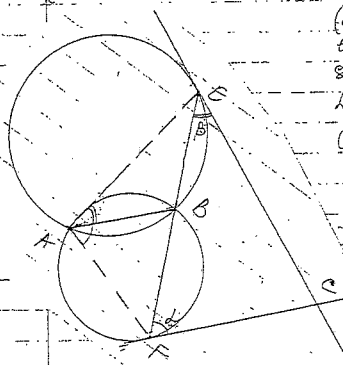
A cross-section = $\frac{8a^2}{3} = \frac{8}{3} \left(\frac{x}{2} \right)^2 = \frac{2x^2}{3}$

$\Delta V = \frac{2x^2}{3} \Delta x$

The solid is symmetrical in its diagonals and the latus rectum of the cross-section equals to $2x$ (say) only for $0 < x \leq \frac{5\sqrt{2}}{2}$, for $x > \frac{5\sqrt{2}}{2}$ the relationship is

$$\begin{aligned} V &= 2 \lim_{\Delta x \rightarrow 0} \sum_{x=0}^{\frac{5\sqrt{2}}{2}} \frac{2x^2}{3} \Delta x \\ &= 2 \int_0^{\frac{5\sqrt{2}}{2}} \frac{2x^2}{3} dx \\ &= \frac{4}{3} \int_0^{\frac{5\sqrt{2}}{2}} x^2 dx \\ &= \frac{4}{3} \left[\frac{x^3}{3} \right]_0^{\frac{5\sqrt{2}}{2}} \\ &= \frac{4}{3} \times \left(\frac{125 \times 2\sqrt{2}}{3 \times 8} - 0 \right) \\ &= \frac{1}{3} \times \frac{125\sqrt{2}}{3} \\ &= \frac{125\sqrt{2}}{9} \text{ units}^3 \end{aligned}$$

(b)



(ii) Let $\widehat{DEF} = d$, then $\widehat{BAF} = d$
(angle between a tangent and a chord equals to the angle in alternate segment subtended by the chord)

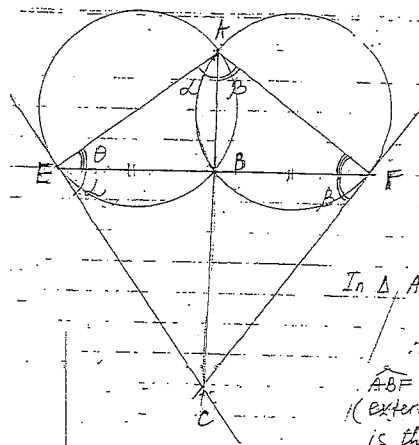
Let $\widehat{CEF} = \beta$, then $\widehat{BAE} = \beta$
(reason as above)

$\widehat{EAF} = d + \beta$ (sum of adjacent angles)

In $\triangle ECF$, $\widehat{ECF} = 180^\circ - (d + \beta)$
(sum of angles in a triangle is 180°)

In quadrilateral $AECF$, $\widehat{ECF} = 180^\circ - \widehat{EAF}$

$\therefore AECF$ is cyclic
(opposite angles are supplementary)



(ii) In equal circles equal chords subtend equal angles.

$\therefore d = \beta$

AB subtends same angle in each circle.

Let this angle be θ

In $\triangle AEF$: $2\theta + 2\theta = 180^\circ$
(angle sum of a triangle is 180°)

$\therefore d + \theta = 90^\circ$

$\widehat{ABF} = d + \theta = 90^\circ$
(exterior angle of a triangle is the sum of two opposite interior angles)

Join BC

In $\triangle EFC$:

$EC = FC$ (sides opposite equal angles are equal, $d = \beta$)

$\therefore \triangle BEC \equiv \triangle BFC$ (SAS: $EB = FB$ (given), $d = \beta$ (proven above), $EC = FC$ (proven above))

$\widehat{EBC} = \widehat{FBC}$ (corresponding angles in congruent triangles are equal)

But $\widehat{EBC} + \widehat{FBC} = \widehat{EBF}$ (adjacent angles)
and $\widehat{EBF} = 180^\circ$ (given: EBF is a straight line)

$\therefore \widehat{FBC} = \frac{180^\circ}{2} = 90^\circ$

$\widehat{ABC} = \widehat{ABF} + \widehat{FBC} = 180^\circ$ (adjacent angles)

$\therefore A, B$ and C are collinear.